

NAYAB KHALID

DATA SCIENCE | MACHINE LEARNING | ARTIFICIAL INTELLIGENCE

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SUMMARY

Final-year Data Science student at COMSATS University Islamabad (CGPA 3.65), specializing in machine learning, computer vision, artificial intelligence and data analytics, with three internships, an authored research paper and an ongoing study in medical AI fairness. Experienced across the full pipeline, from data cleaning through model development and evaluation, with a focus on rigorous measurement and systems that perform reliably for the populations they serve. Committed to building impactful data solutions and advancing diversity within the technology sector.

TECHNICAL SKILLS

Programming: Python, Java, R	AI & Machine Learning: Machine Learning, TensorFlow, AI, NLP, Computer Vision	Web & App Development: React, Next.js, Node.js
Data Science: Pandas, NumPy, Scikit-learn, Jupyter Notebook, Power BI, Tableau, Matplotlib, Seaborn	Cloud & DevOps: AWS, Docker, Jenkins, Git, GitHub	Databases & Big Data: SQL, MongoDB, Hadoop, Spark, Hive

PROFESSIONAL EXPERIENCE

CadetX — Data & AI Intern	Aug 2026 – Present
- Building an NLP and forecasting pipeline over job-posting data to surface emerging skill trends and predict hiring demand. - Collaborating with a four-person team in Python, coordinating development through GitHub to consolidate the analysis into a unified hiring intelligence system.	
FlyRank AI — Machine Learning Intern	Jul 2026 – Aug 2026
- Completed the Refresh and Content Opportunity Scoring capstone across 8 assignment notebooks: framed the research question, verified the data contract against a 9.8M-row warehouse release (30K pages, 44 columns, 32 clients), and built a transparent staleness and visibility baseline before modelling. - Authored a detailed research paper based on the capstone, documenting the methodology, client-grouped validation strategy, and model performance improvements.	
Canteen Stores Department (CSD) — IT Intern	Jul 2025 – Aug 2025
- Analyzed COVID-19's impact on 10K+ employee responses across 4 sectors in Python and Scikit-learn; compared 5 classifiers (Random Forest best on F1), forecast 6 months forward and delivered an interactive Tableau dashboard. - Built OpenCV computer vision projects including Invisibility Cloak and Air Drawing. - Studied CSD's data center end to end, covering infrastructure, systems operations and security, to understand how the organization's IT environment fits together.	

EDUCATION

Bachelor of Science in Data Science	Feb 2023 – Jan 2027
COMSATS University Islamabad Current CGPA: 3.65 Expected graduation: January 2027	
FSc, Pre-Engineering	Sep 2020 – Jul 2022
Punjab Group of Colleges, Ayub Park Campus Grade: A1	
Matric, Computer Science	May 2017 – Feb 2020
Saint Anne's Presentation Convent High School Grade: A+	

RESEARCH PAPERS

Adding Darker Skin Tone Images Does Not Improve Fairness in Melanoma Detection

Co-authored with Noor Ul Huda Batool · COMSATS University Islamabad · 2026

Three-phase controlled study on a MobileNetV2 classifier across Kaggle dermoscopy and the Diverse Dermatology Images set, with skin tone measured by Individual Typology Angle and validated against expert Fitzpatrick labels. Adding 82 authentic darker-skin training images lowered dark-skin specificity from 81.4% to 64.4%, and a matched lighter-skin control degraded it too, identifying domain mismatch between dermoscopic crops and clinical photographs, rather than representation, as the bottleneck.

Refresh & Content Opportunity Scoring: Ranking Pages for Human Review Under Limited Capacity

FlyRank machine learning internship capstone · 2026 · nayab-khalid.github.io/FlyRank-AI-Internship

Binary classification over 30K content pages from 32 pseudonymised clients, with the data contract verified against a 9.8M-row warehouse release and client-grouped validation so no client appeared in both training and test. The corrected model reached 0.596 Average Precision and 0.70 Precision@50 against a 0.489 and 0.30 baseline, with both the leaked and corrected result sets reported in full.

PROJECTS

- **SkinSpectrum** — TensorFlow/Keras, MobileNetV2, OpenCV, React Native, MongoDB, AWS | 2025–Present
Analyzes face photos for 9 skin conditions and melanoma risk using MobileNetV2 (ISIC + HAM10000); includes an AI assistant, product hub, progress tracking and community on a MongoDB Atlas + AWS backend.
- **Netflix Dataset Analysis** — Hadoop, HDFS, PySpark, Hive, HBase, Parquet, D3.js | 2026
8,807 titles through HDFS → PySpark → Hive pipeline, visualised in a cross-filtering D3.js dashboard.
- **Distributed Emotion Detector** — Python, MPI (mpi4py), NLP, Flask | 2026
Real-time emotion classification distributed across 5 MPI processes with a lexicon-based NLP engine, served via Flask.
- **COVID-19 Sector Impact Analysis** — Python, Pandas, Scikit-learn, Tableau | 2025
10K+ employee responses; 5-model comparison and 6-month forecast, delivered as an interactive Tableau dashboard.
- **FocusFlow** — React.js, Next.js, Node.js, Docker, Jenkins, Kubernetes, AWS | 2025
Student productivity app deployed via Docker on AWS EC2, with a Jenkins CI/CD pipeline and Kubernetes orchestration.
- **Hospital Management & Analytics** — Power BI, Tableau, Star Schema | 2025
Star-schema warehouse over 11,400+ records; Power BI and Tableau dashboards tracking revenue, workload and treatment patterns.
- **Football Match Outcome Prediction** — R, randomForest, e1071, mice | 2023
44,557 matches (1872–2023): imputation, then 5 models (LR, Logistic, RF, SVM, NB) compared on goal difference and outcome.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — Stanford University
- Applied Social Network Analysis in Python — University of Michigan
- CCNA: Introduction to Networks — Cisco
- Building with the Claude API — Anthropic
- Model Context Protocol: Advanced Topics — Anthropic
- AI Fluency: Framework & Foundations — Anthropic
- Databases and SQL for Data Science — IBM
- Share Data Through the Art of Visualization — Google
- British Airways Data Science Job Simulation — Forage
- BCG GenAI Job Simulation — Forage

AWARDS & VOLUNTEERING

- **Merit Certificates** — 2nd Position in Class, GPA 3.80 (2023) · 3rd Position in Class, GPA 3.71 (2025)
- **Media Volunteer, Photography & Videography** — COMSATS University Islamabad (2024)
- **Registered Volunteer, Alkhidmat Foundation** — Community Service (2026)

LANGUAGES & INTERESTS

Languages: English — Professional proficiency · Urdu — Native

Interests: Photography, Reading, Cinema, Music, Editing, Journaling